LIMITS OF NONEMPTY QUASI-COMPACT SCHEMES ARE NONEMPTY

THE STACKS PROJECT AUTHORS

Editorial note. No omissions, hints or 'immediate' steps in the counted span.

Allowed standard prerequisites (editorial).

- Directed colimits of rings. Algebra, filtered colimits of rings

Source reference. Original lemma 01YX: sources/stacks-chow/tags.txt, source lines 2572–2572. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $S = \lim S_i$ be the limit of a directed inverse system of schemes with affine transition morphisms (Lemma 01YX). If all the schemes S_i are nonempty and quasi-compact, then the limit $S = \lim_i S_i$ is nonempty.*

Proof. Choose $0 \in I$. Note that I is nonempty as the limit is directed. Choose an affine open covering $S_0 = \bigcup_{j=1,\ldots,m} U_j$. Since I is directed there exists a $j \in \{1, \ldots, m\}$ such that $f_{i0}^{-1}(U_j) \neq \emptyset$ for all $i \geq 0$. Hence $\lim_{i \geq 0} f_{i0}^{-1}(U_j)$ is not empty since a directed colimit of nonzero rings is nonzero (because $1 \neq 0$). As $\lim_{i \geq 0} f_{i0}^{-1}(U_j)$ is an open subscheme of the limit we win. □

Date: stacks-project:limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.